Investigation: FORJAS TAURUS SA (CVM code 6173), fiscal year 2011, as it was on 2013-04-30

Statistical signal based on public financial statements. Not an accusation, not investment advice.

Company

Sector: Metalurgia e Siderurgia (origin: FCA); group: non-financial.
nenhum código de negociação no FCA entregue até a data (nos dados abertos, o campo só vem preenchido a partir do FCA de 2018)

Document used

2011 DFP, version 1, filed on 2012-03-27; origin: open data; type: consolidated; original scale: MIL.
Versions known on the date: 1 (2012-03-27).
Document at the CVM: http://www.rad.cvm.gov.br/ENETCONSULTA/frmDownloadDocumento.aspx?CodigoInstituicao=1&NumeroSequencialDocumento=15473

Possible manipulation (engine v1, study 1)

Beneish M-score
- Value: −2.498 (index)
- Alert by the study 1 rule: no
- Rule: M > −1.78 (pre-registration, section 4)
- Percentile in sector and year: 38 (n = 17)
- Percentile in group and year: 51 (n = 363)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

DSRI, days' sales in receivables index
- Value: 0.985 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 37 (n = 23)
- Percentile in group and year: 39 (n = 457)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

GMI, gross margin index
- Value: 1.071 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 46 (n = 23)
- Percentile in group and year: 69 (n = 467)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

AQI, asset quality index
- Value: 1.210 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 80 (n = 23)
- Percentile in group and year: 84 (n = 462)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

SGI, sales growth index
- Value: 1.015 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 37 (n = 23)

Percentile in group and year: 27 (n = 468)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DEPI, depreciation index
Value: 0.960 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 21 (n = 17)
Percentile in group and year: 41 (n = 377)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

SGAI, SG&A expenses index
Value: 0.945 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 28 (n = 23)
Percentile in group and year: 33 (n = 461)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

TATA, total accruals to total assets
Value: −0.005 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 67 (n = 23)
Percentile in group and year: 67 (n = 484)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

LVGI, leverage index
Value: 1.371 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 93 (n = 23)
Percentile in group and year: 89 (n = 476)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DPO jump
Value: 4.8 days
Alert by the study 1 rule: no
Rule: DPO change in the year's top decile among non-financial companies (≥ 90th percentile of computable values on the date: 34.2 days)
Percentile in sector and year: 57 (n = 22)
Percentile in group and year: 61 (n = 441)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium
Alternative explanations: Bargaining power with suppliers and a change in purchase mix (decision of 2026-11-07)

Implied interest rate minus average CDI
Value: 13.10 p.p.
Alert by the study 1 rule: yes
Rule: implied interest rate above the year's average CDI rate + 10 p.p. (pre-registration, section 4)
Percentile in sector and year: 67 (n = 23)
Percentile in group and year: 71 (n = 429)
The company's own history: fewer than 5 previous fiscal years with a value

Confidence in the data: medium
Alternative explanations: Expensive debt, exchange differences and lease interest inside finance costs (decision of 2026-11-07)

Accruals to average total assets
Value: −0.039 (index)
Alert by the study 1 rule: does not trigger alerts
Percentile in sector and year: 41 (n = 23)
Percentile in group and year: 52 (n = 484)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

Financial risk (context only)

Altman Z'' (financial risk, context only)
Value: 2.996 (index)
Components: X1 = 0.252; X2 = 0.096; X3 = 0.090; X4 = 0.406 (indices)
Percentile in sector and year: 48 (n = 24)
The company's own history: fewer than 5 previous fiscal years with a value
- No cut-off zone: the available source does not define zones for Z''
- Coefficients estimated outside Brazil; bankruptcy models lose validity over time (Pereira and Martins 2015, p. 25)

Financial quality (context only)

Piotroski F (financial quality, context only)
Value: F = 2/9 (8 computable)
Signals:
- Return on assets (ROA): 1
- Operating cash flow on assets (CFO): 1
- Change in ROA: 0
- Cash above earnings (ACCRUAL): 0
- Change in leverage: 0
- Change in current ratio: 0
- No equity issuance (EQ_OFFER): —
- Change in gross margin: 0
- Change in asset turnover: 0
- No equity issuance (EQ_OFFER), from the cash flow statement: 6.03 lines of the frozen "emissões" pattern (reconnaissance, part 4.2), only those about shares or capital, as the article defines it ("did not issue common equity", p. 9); adaptation: preferred shares also count, because in Brazil they are equity, and the line is identified by text in the cash flow statement

Changes

Highlighted in the following year's comparative: 3 (threshold of R$11.3 million, 1% of total assets of the version used). Values in R$ million:
DFC (indirect method) 6.01 Caixa Líquido Atividades Operacionais: 45.3 in the version used (v1, filed on 2012-03-27); 78.6 in the next DFP's comparative (v3, filed on 2013-04-24); difference of 33.3. Different value in the following year's comparative (restatement or reclassification).
BPA 1 Ativo Total: 1,126.7 in the version used (v1, filed on 2012-03-27); 1,114.3 in the next DFP's comparative (v3, filed on 2013-04-24); difference of −12.3. Different value in the following year's comparative (restatement or reclassification).
BPP 2.02 Passivo Não Circulante: 335.9 in the version used (v1, filed on 2012-03-27); 323.5 in the next DFP's comparative (v3, filed on 2013-04-24); difference of −12.3. Different value in the following year's comparative (restatement or reclassification).

Highlighted: differences in totals and key lines (total assets, current liabilities, non-current liabilities, equity, net revenue, net income and operating cash flow) above 1% of total assets of the version used. The other differences (reclassifications between lines and smaller differences) stay in the full list.
Full list: 37 accounts above the R$1,000 tolerance (group a, totals and key lines: 3; group b, reclassifications: 34), in the annex at the end.

What was known on the date

- FCA 2011, v1: 2011-03-25
- DFP 2010, v1: 2011-03-30
- DFP 2010, v2: 2011-03-30
- DFP 2010, v3: 2011-04-12
- FCA 2011, v2: 2011-05-31
- FCA 2011, v3: 2011-08-12
- DFP 2011, v1: 2012-03-27
- DFP 2012, v1: 2013-03-26
- DFP 2012, v2: 2013-04-04
- DFP 2012, v3: 2013-04-24

DFP and FCA filed with the CVM by the reference date; ITR and material facts come in later phases.

Data quality

53 applicable checks; 2 failed.

- Cash flow closing balance = cash and cash equivalents on the balance sheet (PENÚLTIMO): difference above the tolerance of R$2,000
- Cash flow opening balance = cash and cash equivalents on the prior balance sheet (ÚLTIMO): difference above the tolerance of R$2,000

Auditor's report and auditor (factual section; no alert)

Auditor's report: Sem Ressalva (unqualified).
Excerpt checked literally: "Em nossa opinião, as demonstrações contábeis individuais referidas no primeiro parágrafo apresentam adequadamente, em todos os aspectos relevantes, a posição patrimonial e financeira da Forjas Taurus S.A. em 31 de dezembro de 2011, o desempenho de suas operações e os seus fluxos de caixa para o exercício findo naquela data, de acordo com as práticas contábeis adotadas no Brasil."
Auditor in the 2010 FCA: KPMG Auditores Independentes.
Auditor in the 2011 FCA: KPMG Auditores Independentes.
Auditor in the 2012 FCA: Ernst & Young Terco Auditores Independentes S/S, KPMG Auditores Independentes.
Auditor changes (one line per change, canonical name by the firm's CVM code):
- 2012: entry of Ernst & Young Terco Auditores Independentes S/S, with no other firm leaving.

Auditor rotation: under CVM Resolution No. 23/2021, art. 31, the same independent auditor may not serve the same client for more than five consecutive fiscal years, with a minimum interval of three fiscal years before being hired again; under art. 31-A, the term is up to ten consecutive fiscal years when the company has a statutory audit committee in permanent operation and the auditor is a firm. The Resolution has been in force since 2021-04-01 (art. 41); before that, the equivalent rule was in arts. 31 and 31-A of CVM Instruction No. 308/1999, revoked by the Resolution (art. 40, I). A change of auditor is a fact, not an alert.

Document excerpts (copied; no alert)

Excerpts copied from the documents, with the page, and the Reference Form table. It is not a signal family: it generates no alert, score or ranking. Absence of mention does not prove absence.
How the reader was tested: 20 held-out cases, a single measurement with the frozen reader (c99c953): precision 0.962, coverage 0.962, literal check on 132 of 132 excerpts, no natural-person name (docs/leitor_medida.json).

Explanatory notes · Related parties

Note: 23 Partes relacionadas (pages 53 to 54)

- [key management personnel] “Remuneração de pessoal-chave da administração” (p. 53, original excerpt in Portuguese)
- [key management personnel] “Em 31 de dezembro de 2011, a remuneração do pessoal-chave da administração, que contempla a” (p. 53, original excerpt in Portuguese)
- [key management personnel] “Direção e o Conselho de Administração da Companhia, totalizou R$ 26.894 (R$ 33.026 em 31 de” (p. 53, original excerpt in Portuguese)
- [key management personnel] “Remuneração de pessoal-chave da administração compreende:” (p. 53, original excerpt in Portuguese)
- [parent company] “2011 2010 2011 2010” (p. 53, original excerpt in Portuguese)
- [key management personnel] “Remuneração e benefícios dos diretores estatutários e Conselho de” (p. 53, original excerpt in Portuguese)
- [key management personnel] “Remuneração de pessoal-chave 11.926 15.112 7.150 7.475” (p. 53, original excerpt in Portuguese)
- [key management personnel] “Total 26.894 33.026 22.118 21.440” (p. 53, original excerpt in Portuguese)
- [key management personnel] “A Companhia não possui políticas de benefícios de remuneração ao pessoal-chave da administração que se caracterizem como: benefícios pós-emprego, benefícios de rescisão de contrato de trabalho, remuneração com base em ações ou outros benefícios de longo prazo.” (p. 53, original excerpt in Portuguese)
- Excerpt omitted; see page 53 of the document
- [key management personnel] “Operações de diretores e pessoal-chave da administração” (p. 53, original excerpt in Portuguese)
- [key management personnel] “Os diretores e pessoal-chave da administração controlam diretamente 0,69% das ações com direito de voto da Companhia.” (p. 53, original excerpt in Portuguese)
- [parent company; subsidiaries] “controlada Taurus Máquinas-Ferramenta Ltda. possui saldo a receber da Wotan Máquinas Ltda., no valor de R$ 34.136. A controlada Taurus Security Ltda. possui contrato de mútuo com a controladora Forjas Taurus S.A. no valor de R$ 219.” (p. 53, original excerpt in Portuguese)
- [parent company; subsidiaries] “A controlada Taurus Máquinas-Ferramenta Ltda. possui contrato de mútuo a pagar com a controladora” (p. 53, original excerpt in Portuguese)
- [subsidiaries; key management personnel] “Os empréstimos e financiamentos estão garantidos por notas promissórias, alienação fiduciária de máquinas e equipamentos, hipoteca de imóveis e aval dos diretores. Os avais concedidos por diretores e os avais concedidos pela Companhia à suas controladas estão demonstrados a seguir:” (p. 54, original excerpt in Portuguese)
- [key management personnel] “Avais de diretores:” (p. 54, original excerpt in Portuguese)
- [key management personnel; subsidiaries] “2011 2010” (p. 54, original excerpt in Portuguese)
- [key management personnel] “Forjas Taurus S.A. - 16.636” (p. 54, original excerpt in Portuguese)
- [subsidiaries] “A Companhia prestou avais às suas controladas, cujos montantes seguem abaixo discriminados:” (p. 54, original excerpt in Portuguese)
- [subsidiaries] “Avais às controladas:” (p. 54, original excerpt in Portuguese)
- [subsidiaries] “Taurus Máquinas-Ferramenta Ltda. 17.391 31.256” (p. 54, original excerpt in Portuguese)
- [subsidiaries] “Taurus Holdings, Inc. 55.002 42.533” (p. 54, original excerpt in Portuguese)
- [subsidiaries] “72.393 73.789” (p. 54, original excerpt in Portuguese)

Fields of item 18 of CPC 05 (R1)

- [amount of the transactions] “Direção e o Conselho de Administração da Companhia, totalizou R$ 26.894 (R$ 33.026 em 31 de” (p. 53, original excerpt in Portuguese)
- [amount of the transactions] “resultado do exercício, cujo montante é de R$ 4.146 em 2011(R$ 6.982 em 2010).” (p. 53, original excerpt in Portuguese)
- [balances] “Algumas pessoas-chave da administração detêm quotas em outras entidades, fazendo com que tenham controle ou influência significativa sobre as políticas financeiras ou

operacionais destas entidades. Os saldos pendentes com estas partes relacionadas são avaliados com base em termos do mercado e estão detalhados abaixo:" (p. 53, original excerpt in Portuguese)
- [balances] "controlada Taurus Máquinas-Ferramenta Ltda. possui saldo a receber da Wotan Máquinas Ltda., no valor de R$ 34.136. A controlada Taurus Security Ltda. possui contrato de mútuo com a controladora Forjas Taurus S.A. no valor de R$ 219." (p. 53, original excerpt in Portuguese)
- [balances] "A controlada Taurus Máquinas-Ferramenta Ltda. possui contrato de mútuo a pagar com a controladora" (p. 53, original excerpt in Portuguese)
- [terms and conditions] "Todos os contratos de mútuo entre as partes relacionadas possuem vencimento no longo prazo." (p. 53, original excerpt in Portuguese)
- [guarantees] "Avais entre partes relacionadas" (p. 54, original excerpt in Portuguese)
- [guarantees] "Os empréstimos e financiamentos estão garantidos por notas promissórias, alienação fiduciária de máquinas e equipamentos, hipoteca de imóveis e aval dos diretores. Os avais concedidos por diretores e os avais concedidos pela Companhia à suas controladas estão demonstrados a seguir:" (p. 54, original excerpt in Portuguese)
- [guarantees] "Avais de diretores:" (p. 54, original excerpt in Portuguese)
- [guarantees] "2011 2010" (p. 54, original excerpt in Portuguese)
- [guarantees] "Forjas Taurus S.A. - 16.636" (p. 54, original excerpt in Portuguese)
- [amount of the transactions; guarantees] "A Companhia prestou avais às suas controladas, cujos montantes seguem abaixo discriminados:" (p. 54, original excerpt in Portuguese)
- [guarantees] "Avais às controladas:" (p. 54, original excerpt in Portuguese)
- [guarantees] "Taurus Máquinas-Ferramenta Ltda. 17.391 31.256" (p. 54, original excerpt in Portuguese)
- [guarantees] "Taurus Holdings, Inc. 55.002 42.533" (p. 54, original excerpt in Portuguese)
- [guarantees] "72.393 73.789" (p. 54, original excerpt in Portuguese)

Explanatory notes · Reverse factoring and forfaiting
The reader found no mention of reverse factoring (risco sacado), forfaiting or supplier agreements in the notes of this DFP. Absence of mention does not prove absence of the operation.

Auditor's report
Type of opinion: unqualified
- [emphasis of matter] "Ênfase Conforme descrito na nota explicativa 3, as demonstrações contábeis individuais foram elaboradas de acordo com as práticas contábeis adotadas no Brasil. No caso da Forjas Taurus S.A. essas práticas diferem da IFRS, aplicável às demonstrações contábeis separadas, somente no que se refere à avaliação dos investimentos em controladas, coligadas e controladas em conjunto pelo método de equivalência patrimonial, enquanto que para fins de IFRS seria custo ou valor justo. Nossa opinião não está ressalvada em função desse assunto." (original excerpt in Portuguese)
- Excerpt omitted; see the auditor's report in the document

Reference Form
The Reference Form for this fiscal year was not available on the date. It is the only version available in the open data; earlier versions are not there.

Limits of this section:
- In real documents, about 22% of the excerpts come out omitted by the name detector, including names of companies outside the registry.
- In tables, header lines (years, column titles) may come out as excerpts; the held-out cases had no tables with multi-line headers.

How the engine was tested

In study 1, engine v1 detected 7 of 13 detectable cases, with 108.3 false alarms per detected case over the whole period and an alert rate of 49.9% among evaluated non-financial company-years (docs/summary.md).

Limits

- No alert is not a clean bill of health. In Dechow, Sloan and Sweeney (1995), the most powerful accruals model rejected 'no manipulation' for only 28.1% of 32 companies sanctioned by the SEC (Table 7, p. 32).
- In study 1, 49.9% of evaluated non-financial company-years (48.3% including financial companies) were in alert, and the number of detections did not beat chance in a descriptive test done after the result: 7 detections in 13 detectable cases, against 7.37 expected by chance without assuming independent years (8.56 assuming independent years).
- Risk models use numbers that may be manipulated (Pereira and Martins 2015, p. 29). That is why the three families are not added up.

Limitations

- Context indicators (Altman Z'' and Piotroski F) were not validated in Brazil by this project and do not trigger alerts

Evidence in the JSON: 39 DFP lines, each with document, version, filing date, account and location.

Engine

motor v1 (estudo 1), branch estudo-1-final, commit 8c9a823; chiocca.v2 (produto, fases B a F1); leitor-f1-1 (commit c99c953). Generated at 2026-11-12T03:40:25+00:00.

Today's information (outside the reference date)

Registry status: ATIVO. Source: CVM registry of listed companies (cad_cia_aberta.csv), in the version downloaded for the project, not on the reference date.
Name in today's registry: TAURUS ARMAS S.A.

Annex: full list of differences in the following year's comparative (R$ million)

(a) DFC (indirect method) 6.01 Caixa Líquido Atividades Operacionais: 45.3 in the version used, 78.6 in the comparative (difference of 33.3)
(a) BPA 1 Ativo Total: 1,126.7 in the version used, 1,114.3 in the comparative (difference of −12.3)
(a) BPP 2.02 Passivo Não Circulante: 335.9 in the version used, 323.5 in the comparative (difference of −12.3)
(b) DFC (indirect method) 6.03.03 Empréstimos Tomados: 322.0 in the version used, −126.5 in the comparative (difference of −448.5)
(b) DFC (indirect method) 6.03.02 Pagamentos Juros s/ Capital Próprio e Dividendos: −15.8 in the version used, 322.0 in the comparative (difference of 337.8)
(b) DFC (indirect method) 6.03.06 Reestruturação Societária: −169.4 in the version used, 0.0 in the comparative (difference of 169.4)
(b) DFC (indirect method) 6.03.05 Ações em Tesouraria: −32.9 in the version used, −169.4 in the comparative (difference of −136.5)
(b) DFC (indirect method) 6.03.04 Pagamentos de Empréstimos: −126.5 in the version used, −32.9 in the comparative (difference of 93.6)
(b) BPA 1.01.01 Caixa e Equivalentes de Caixa: 74.8 in the version used, 162.2 in the comparative (difference of 87.5)
(b) BPA 1.01.02 Aplicações Financeiras: 87.5 in the version used, 0.0 in the comparative (difference of −87.5)
(b) BPA 1.01.02.02 Aplicações Financeiras Avaliadas ao Custo Amortizado: 87.5 in the version used, 0.0 in the comparative (difference of −87.5)
(b) BPA 1.01.02.02.01 Títulos Mantidos até o Vencimento: 87.5 in the version used, 0.0 in the comparative (difference of −87.5)
(b) BPP 2.01.05.02.07 Outras Obrigações: 40.3 in the version used, 0.0 in the comparative (difference of −40.3)

(b) DFC (indirect method) 6.01.01.07 Provisão de Juros s/ Empréstimos e Financiamentos: 38.9 in the version used, −0.9 in the comparative (difference of −39.8)
(b) DFC (indirect method) 6.01.01.10 Variação Cambial Líquida de Investimento Exterior: 0.0 in the version used, 38.9 in the comparative (difference of 38.9)
(b) DFC (indirect method) 6.03 Caixa Líquido Atividades de Financiamento: −23.4 in the version used, −56.7 in the comparative (difference of −33.3)
(b) DFC (indirect method) 6.03.10 Débitos com Empresas Ligadas: 0.0 in the version used, −33.3 in the comparative (difference of −33.3)
(b) DFC (indirect method) 6.01.01.05 Imp. Renda e Contr. Social Diferidos: −33.5 in the version used, −1.9 in the comparative (difference of 31.6)
(b) DFC (indirect method) 6.01.03 Outros: −28.9 in the version used, −6.1 in the comparative (difference of 22.9)
(b) DFC (indirect method) 6.01.01.01 Lucro Líquido do Exercício: 37.3 in the version used, 17.6 in the comparative (difference of −19.7)
(b) DFC (indirect method) 6.03.01 Participação dos Minoritários: 0.0 in the version used, −15.8 in the comparative (difference of −15.8)
(b) DFC (indirect method) 6.01.01 Caixa Gerado nas Operações: 74.9 in the version used, 88.7 in the comparative (difference of 13.9)
(b) BPA 1.02 Ativo Não Circulante: 376.6 in the version used, 364.3 in the comparative (difference of −12.3)
(b) BPA 1.02.01 Ativo Realizável a Longo Prazo: 62.9 in the version used, 50.6 in the comparative (difference of −12.3)
(b) BPA 1.02.01.06 Tributos Diferidos: 56.1 in the version used, 43.8 in the comparative (difference of −12.3)
(b) BPA 1.02.01.06.01 Imposto de Renda e Contribuição Social Diferidos: 56.1 in the version used, 43.8 in the comparative (difference of −12.3)
(b) BPP 2 Passivo Total: 1,126.7 in the version used, 1,114.3 in the comparative (difference of −12.3)
(b) BPP 2.02.03 Tributos Diferidos: 18.4 in the version used, 6.1 in the comparative (difference of −12.3)
(b) BPP 2.02.03.01 Imposto de Renda e Contribuição Social Diferidos: 18.4 in the version used, 6.1 in the comparative (difference of −12.3)
(b) DFC (indirect method) 6.01.01.06 Equivalência Patrimonial: −1.9 in the version used, 3.7 in the comparative (difference of 5.6)
(b) DRE 3.04.04 Outras Receitas Operacionais: 0.0 in the version used, 4.3 in the comparative (difference of 4.3)
(b) DRE 3.04.05 Outras Despesas Operacionais: −9.0 in the version used, −13.3 in the comparative (difference of −4.3)
(b) DFC (indirect method) 6.01.01.08 Provisão Instrumentos Financeiros Derivativos: 3.7 in the version used, 0.0 in the comparative (difference of −3.7)
(b) DFC (indirect method) 6.01.02 Variações nos Ativos e Passivos: −0.6 in the version used, −4.0 in the comparative (difference of −3.5)
(b) DFC (indirect method) 6.01.02.05 Aumento (Redução) em Contas a Pagar e Provisões: 55.0 in the version used, 51.5 in the comparative (difference of −3.5)
(b) DFC (indirect method) 6.03.11 Impostos e Taxas a Recolher: 0.0 in the version used, −0.8 in the comparative (difference of −0.8)
(b) BPP 2.01.05.02.05 Saques Cambiais: 39.4 in the version used, 39.6 in the comparative (difference of 0.2)